

Prompt (Template 3)

A group of {71} students has various hobbies. {4} like to read, {6} like to play basketball, and the rest like to either play music or write. How many like to play music if the number that like to write is {10} times the number that prefer playing basketball?

First Output (Qwen2.5-Math-7B-Instruct) (Incorrect)

To determine how many students like to play music, we need to follow these steps: ...

$$\text{Total} = 4 + 6 + 60 = 64$$

$$\text{Number of students who like to play music} = 71 - 64 = 7$$

Thus, the number of students who like to play music is 7.

Second Output (Incorrect)

To determine how many students like to play music, we start by identifying the number of students who like to write. According to the problem, the number of students who like to write is 10 times the number of students who like to play basketball. ...

$$71 - 4 - 60 = 7$$

Thus, the number of students who like to play music is 7.

Third Output (Correct)

To determine how many students like to play music, we start by identifying the number of students who like to write. ... Next, let m represent the number of students who like to play music. Then:

$$4 + 6 + m + 60 = 71$$

$$70 + m = 71$$

$$m = 1$$

Therefore, the number of students who like to play music is 1.

Fourth Output (Incorrect)

To determine how many students like to play music, we need to follow these steps: ...

$$\text{Total} = 4 + 6 + 60 = 66$$

$$\text{Number of students who like to play music} = 71 - 66 = 5$$

Thus, the number of students who like to play music is 5.

Fifth Output (Incorrect)

To determine how many students like to play music, we start by identifying the number of students who like to write. ...

$$4 + 6 + 60 = 66$$

Since the total number of students is 71, the number of students who like to play music is:

$$71 - 66 = 5$$

Thus, the number of students who like to play music is:

5

Sixth Output (Incorrect)

To determine how many students like to play music, we need to follow these steps: ...

$$\text{Total} = 4 + 6 + 60 = 60$$

$$\text{Number of students who like to play music} = 71 - 60 = 11$$

Thus, the number of students who like to play music is 11.

Seventh Output (Correct)

To determine how many students like to play music, we start by identifying the number of students who like to write. ...

$$71 - (4 + 6 + 60) = 71 - 70 = 1$$

Thus, the number of students who like to play music is 1.

Eighth Output (Incorrect)

To determine how many students like to play music, we need to follow these steps: ...

$$\text{Total} = 4 + 6 + 60 = 66$$

$$\text{Number of students who like to play music} = 71 - 66 = 5$$

Thus, the number of students who like to play music is 5.

Answer:1

Figure 18: Eight independently generated outputs from Qwen2.5-Math-7B-Instruct evaluated on Template 3. Error type: wrong addition/subtraction.

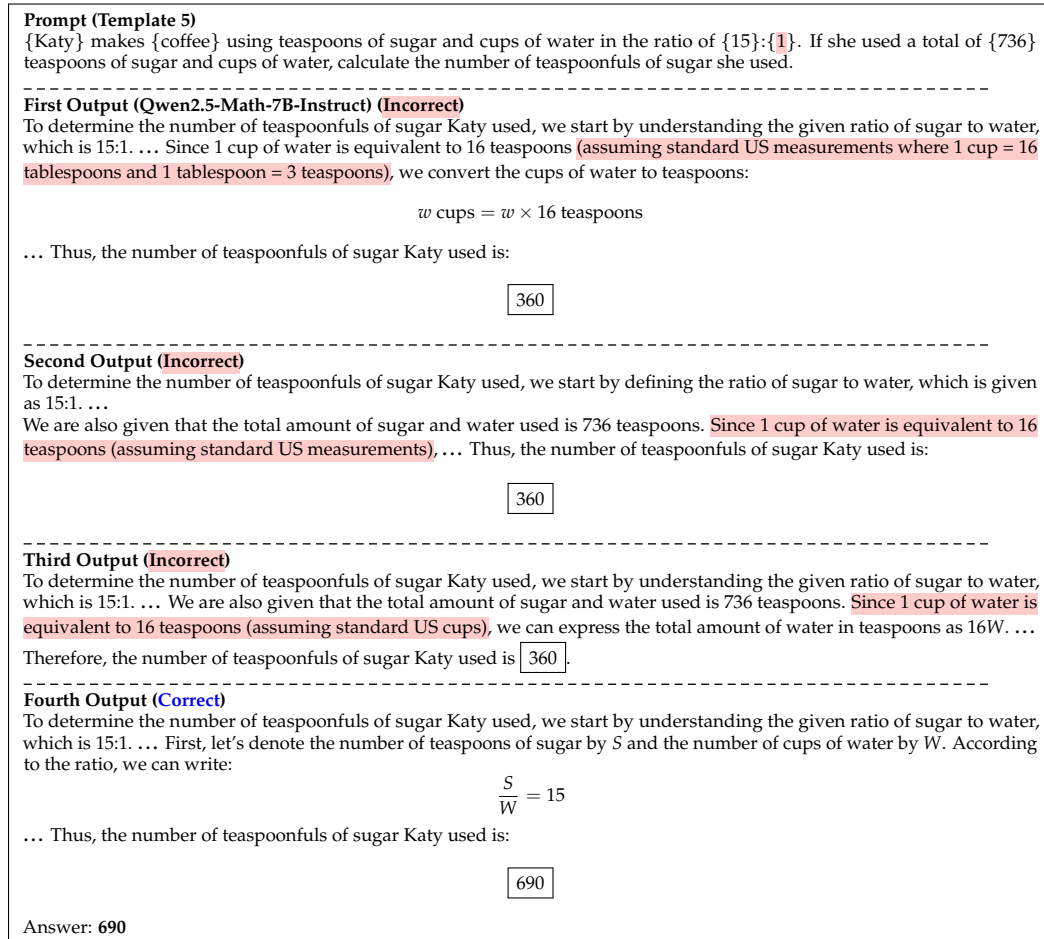


Figure 19: First four of eight independently generated outputs from Qwen2.5-Math-7B-Instruct evaluated on Template 5. Error type: unjustified unit conversion.

Prompt (Template 6) {John} has {15} {hectares} of {pineapples} field. There are {47} {pineapples} per {hectare}. {John} can harvest his {pineapples} every {2} months. How many {pineapples} can {John} harvest within a year?
First Output (Qwen2.5-Math-7B-Instruct) (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... Total number of pineapples = 15 hectares $\times$ 47 pineapples/hectare = 655 pineapples Therefore, the total number of pineapples John can harvest within a year is 3930.
Second Output (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... Total number of pineapples = 15 hectares $\times$ 47 pineapples/hectare = 695 pineapples Therefore, the total number of pineapples John can harvest within a year is 4170.
Third Output (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... 15 hectares $\times$ 47 pineapples/hectare = 655 pineapples Therefore, the number of pineapples John can harvest within a year is 3930.
Fourth Output (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... 15 hectares $\times$ 47 pineapples/hectare = 655 pineapples Therefore, the total number of pineapples John can harvest within a year is: 3930
Fifth Output (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... Total number of pineapples = 15 hectares $\times$ 47 pineapples/hectare = 695 pineapples Therefore, the number of pineapples John can harvest within a year is 4170.
Sixth Output (Incorrect) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... Total number of pineapples = 15 hectares $\times$ 47 pineapples/hectare = 655 pineapples Therefore, the total number of pineapples John can harvest within a year is 3930.
Seventh Output (Correct) To determine how many pineapples John can harvest within a year, we need to follow these steps: ... 15 hectares $\times$ 47 pineapples/hectare = 705 pineapples Therefore, the number of pineapples John can harvest within a year is 4230. Answer: 4230

Figure 20: Seven of eight independently generated outputs from Qwen2.5-Math-7B-Instruct evaluated on Template 6. Error type: wrong multiplication.